Chapter 18

Dynamics of the Southern Ocean Circulation

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1. INTRODUCTION

The existence of a circumpolar oceanic channel in the latitude band of Drake Passage has profound implications for the global ocean circulation and climate. The Drake Passage gap permits the existence of the Antarctic Circumpolar Current (ACC), a system of ocean currents that flows from west to east along a roughly 25,000-km-long path circling Antarctica (Figure 18.1). The strong eastward flow of the ACC has several important implications for the global ocean circulation and its influence on regional and global climate. It promotes exchange between the three major ocean basins, allowing transmission of climate signals and smoothing out zonal differences in water properties, yet inhibits north–south exchange and isolates Antarctica from the warm waters to the north. The interbasin connection (see Chapter 19) allows the establishment of a global-scale overturning circulation, which transports heat, moisture, and carbon dioxide around the globe and strongly influences the Earth's climate. The eastward geostrophic flow of the ACC is associated with steeply sloping density surfaces, which shoal to the south across the current and expose dense waters to the surface in the high-latitude Southern Ocean (Figure 18.2). Where the dense waters outcrop at the sea surface, they exchange heat, moisture, and gases like oxygen and carbon dioxide with the atmosphere. In this sense, the Southern Ocean acts as a valve regulating exchange between the deep and shallow layers of the global ocean. Upwelling of deep water returns nutrients and carbon to the upper ocean, while sinking of waters from the surface transfers oxygen and anthropogenic carbon dioxide (CO_2) into the ocean interior. As a result of the

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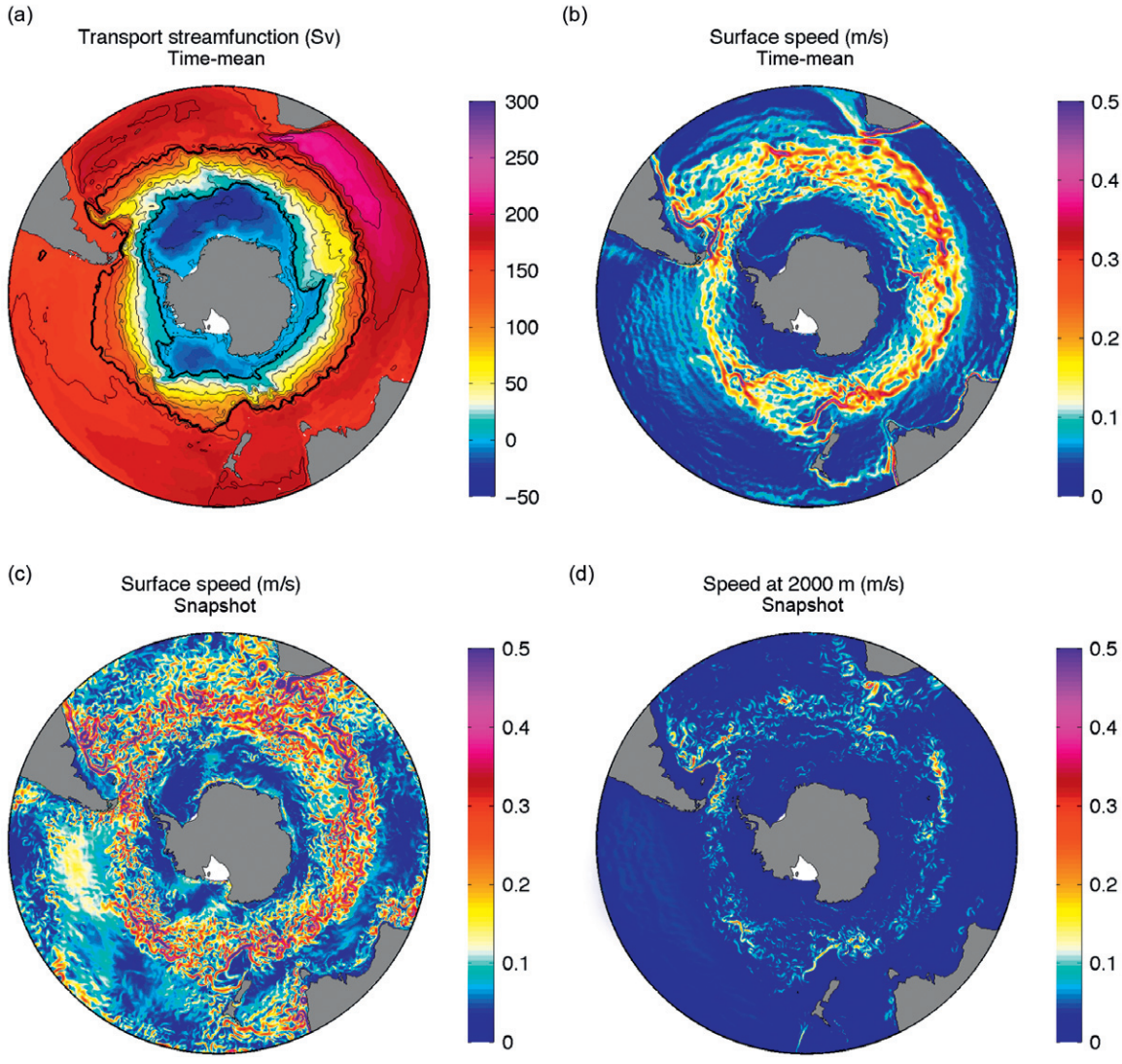

FIGURE 18.1 Four views of the Antarctic Circumpolar Current based on the Southern Ocean State Estimate of Mazloff et al. (2010). (a) Five-year mean (2005–2009) vertically integrated transport streamfunction (in Sv); bold contours indicate the northern and southernmost streamlines to pass through Drake Passage. (b) Five-year mean surface velocity (m s^{-1}). (c) Five-day mean snapshot of surface velocity in mid-January 2005. (d) Five-day mean velocity at 2000 m depth. *Figure courtesy of M. Mazloff and B. Peña Molino.*

vigorous overturning circulation, the Southern Ocean stores large amounts of heat and anthropogenic CO_2 (see Chapter 30). For example, 40% of the total ocean inventory of anthropogenic CO_2 entered the ocean south of 40°S (Khatiwala et al., 2009, 2013).

Understanding of the Southern Ocean circulation has been slowed by a scarcity of observations. The measurements collected during the World Ocean Circulation Experiment (WOCE) were a significant achievement. The WOCE hydrographic program, for example, provided the first circumpolar survey of the Southern Ocean since the epic voyages of the *Eltanin* in the late 1960s and early 1970s. But over the last decade or so, even more remarkable advances have been made in our capacity to observe the Southern Ocean (Chapter 3). More than 150,000 temperature and salinity profiles to 2000 m depth have been collected south of 40°S since the Argo profiling float program began in 2003. Oceanographic sensors deployed on elephant seals for biology studies have provided more than 50,000 temperature and salinity profiles, many of them from the sea ice zone in winter where few measurements have been made with traditional techniques. The satellite altimeter record now exceeds two decades in length, allowing an unprecedented year-round, circumpolar view of the Southern Ocean circulation (Chapter 4). Repeated hydrographic sections have quantified the evolution of a wide range of ocean properties throughout the full ocean depth. Direct observations of diapycnal mixing are now possible (Chapter 7). Taken together, these advances have revolutionized our ability to observe and understand the Southern Ocean.

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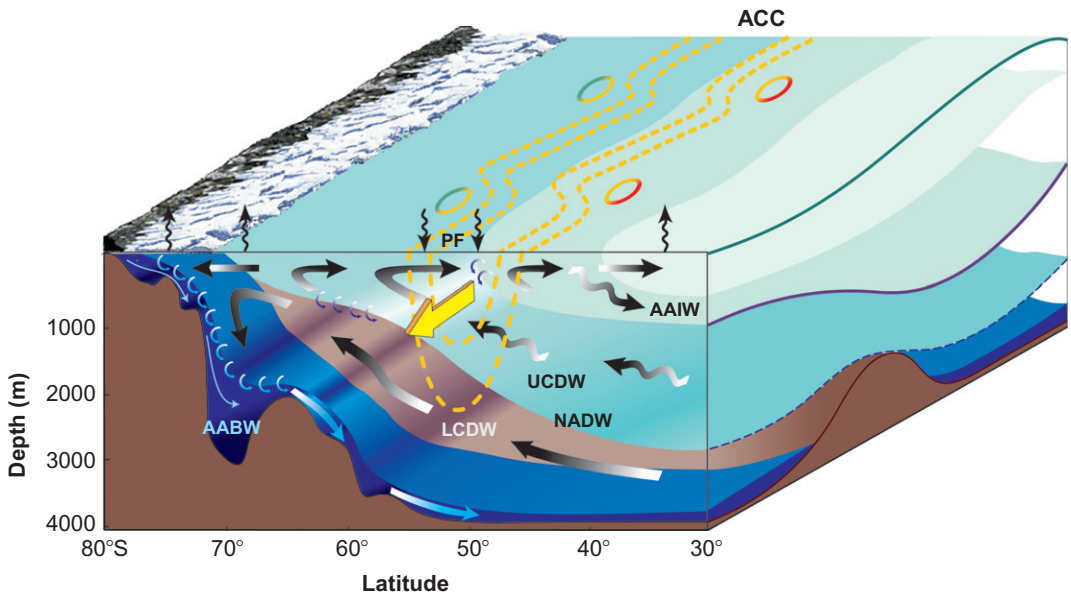

FIGURE 18.2 A schematic view of the Southern Ocean circulation. The heavy yellow arrow and dashed yellow lines denote the eastward flow of the Antarctic Circumpolar Current. The overturning circulation is indicated by dark arrows, with wavy arrows intended to represent transport by eddies. The meridional transport is largely along layers of constant density, represented by the colored surfaces. Vertical curly arrows at the sea surface indicate air–sea buoyancy exchange (upward arrows mean a buoyancy loss by the ocean). Small light arrows show diapycnal mixing. The east–west section on the right face of the diagram illustrates the surface and isopycnal tilts in relation to bottom topography (brown) associated with interfacial and bottom form stress. *From Olbers and Visbeck (2005), adapted from Speer et al. (2000).*

Progress in modeling the Southern Ocean has been equally dramatic. Near the beginning of WOCE, the Fine Resolution Antarctic Model (FRAM) was completed, a regional model of the Southern Ocean with (1/4° latitude) × (1/2° longitude) resolution (an achievement so remarkable at that time that a large atlas was published with results from a snapshot of a single model run (Webb et al., 1991)). The FRAM model, which took 3 years of integration time to complete 6 years of simulation, was the first numerical representation of the Southern Ocean to include mesoscale features like eddies, fronts, and meanders. Global models are now being run for decades with a spatial resolution of 1/10° (Chapter 20). Even more importantly for improving understanding, advances in computing power have allowed many runs to be completed at high resolution, allowing testing of the sensitivity of the circulation to changes in forcing and to model parameters. Eddies play a particularly important role in the Southern Ocean, and the development of high-resolution models has allowed the dynamical significance of eddies to be explored in detail for the first time.

Several contemporary reviews have done a comprehensive job of summarizing progress in understanding the Southern Ocean circulation (e.g., Olbers et al., 2004; Meredith et al., 2011; Marshall and Speer, 2012; chapter 16 of Olbers et al., 2012). Here we aim to complement these studies by summarizing recent advances in the understanding of Southern Ocean dynamics gained from observations, theory, and modeling. We begin with a brief review of what had been learned about Southern Ocean dynamics during WOCE, as described in the first edition of the "WOCE book" (Rintoul et al., 2001) (Section 2). In Section 3, advances in understanding the structure and dynamics of the ACC are summarized. A common element of these recent developments is an increasing recognition of the central role played by small- and mesoscale processes, often localized by topography, in the dynamics and climatic sensitivity of the Southern Ocean circulation. Recent progress in understanding the dynamics of the Southern Ocean overturning circulation is reviewed in Section 4. New observations have provided time series of sufficient length to provide insight into the nature and causes of Southern Ocean change, as discussed in Section 5. In Section 6, the main conclusions of this chapter are presented, and a brief outlook on future challenges put forward.

2. PROGRESS IN UNDERSTANDING SOUTHERN OCEAN DYNAMICS DURING WOCE (1990–2002)

Rintoul et al. (2001) reviewed progress made during the WOCE decade in understanding the ACC system. New observations and advances in theory and numerical modeling during WOCE allowed the synthesis of a "dynamical recipe" for the circulation of the Southern Ocean. The key ingredients include wind and buoyancy forcing, eddies, stratification, and topography. Both wind and buoyancy forcing (including ocean–ice interaction) act to tilt isopycnals so they shoal to the south, producing the strong eastward geostrophic flow of the ACC. When

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the tilt is sufficiently large, baroclinic instability taps the available potential energy stored in the sloping isopycnals to produce an energetic eddy field. Eddies play a number of critical roles in the ACC system. They carry heat poleward, to balance oceanic heat loss to the atmosphere at high latitude. Poleward eddy heat flux is associated with vertical transport of horizontal momentum. The vertical momentum transport (or interfacial form stress) establishes deep geostrophic flows balanced by pressure gradients across bathymetric features, resulting in bottom form stress that opposes the wind stress at the sea surface. Divergence of the eddy-induced interfacial form stress supports meridional transport in layers not blocked by topography, where no time-mean meridional geostrophic flow is possible. The dynamics of the ACC and the meridional overturning circulation are therefore fundamentally linked, as illustrated schematically in Figure 18.2.

Interaction of the flow with topography is also central to the vorticity balance, with the dominant balance in the barotropic vorticity equation being between bottom pressure torque and advection of planetary vorticity (in contrast to the flat-bottom Sverdrup balance between wind stress curl and advection of planetary vorticity that holds in most other parts of the ocean) (Hughes and de Cuevas, 2001). Bottom topography has a greater influence in the Southern Ocean because of the relatively weak stratification and deep-reaching nature of the flow. The stratification is weak, but not zero, and partially shields the flow from the effects of the bathymetry: barotropic models with realistic topography, forcing, and friction typically have very weak circumpolar flow (e.g., Olbers et al., 2004).

The Southern Ocean overturning circulation consists of two cells (Figure 18.2). Circumpolar Deep Water (CDW) spreads poleward and shoals, ultimately outcropping at the sea surface. The relatively dense Lower CDW upwells to the south of the ACC, where interactions with the atmosphere and sea ice convert the upwelled water to dense Antarctic Bottom Water (AABW), forming the lower cell of the overturning circulation. Relatively light Upper CDW outcrops within the ACC belt, where it is driven equatorward in the Ekman layer and converted to less dense intermediate waters by heat and freshwater gain from the atmosphere, forming the upper cell. In this way, the Southern Ocean connects the upper and lower layers of the ocean and allows a global-scale overturning circulation to exist even in the limit of very weak diapycnal mixing in the ocean interior (e.g., Speer et al., 2000).

While the observations and theoretical advances made during the WOCE era led to much deeper understanding of the Southern Ocean circulation, understanding of the interactions between the dynamical ingredients identified above remained incomplete. In particular, the sensitivity of the ACC and overturning circulation to changes in wind and buoyancy forcing was poorly understood. Observational evidence for change in the Southern Ocean was sparse due to the lack of long records. In the following, we review progress made in the last decade in understanding the dynamics of the Southern Ocean circulation, much of it built on the foundation laid during WOCE.

3. THE ANTARCTIC CIRCUMPOLAR CURRENT (ACC)

3.1. Structure of the ACC

The eastward flow of the ACC is concentrated in narrow fronts or jets, as first recognized by Sir George Deacon in the 1930s. For example, Orsi et al. (1995) identified three circumpolar fronts: from north to south, the Subantarctic Front (SAF), Polar Front (PF), and southern ACC front, as well as a fourth feature they called the "southern boundary of the ACC," coincident with the southern limit of the flow through Drake Passage and not always associated with a current core. Each of the fronts was found to coincide with a water mass boundary (and hence could be reliably identified in hydrographic data), as well as a maximum in current speed (and hence associated with enhanced horizontal gradients in density and sea surface height).

This classical view of the ACC fronts is not easily recognizable in Figure 18.1, which shows four perspectives on the structure of the ACC derived from the Southern Ocean State Estimate (SOSE; Mazloff et al., 2010). SOSE assimilates a large volume of observations (e.g., altimetry, moorings, and temperature and salinity profiles collected by floats, ships, and seals) into an eddy-permitting (1/6° resolution) general circulation model (Chapter 21 discusses ocean state estimates). The resulting flow field can be thought of as a dynamically consistent interpolation of sparse observations. The circumpolar flow of the ACC clearly dominates the vertically integrated circulation south of 25°S. While the zonal continuity of the ACC is the most striking feature in Figure 18.1a, it is also clear that the path and width of the ACC varies significantly around the circumpolar belt. Cyclonic gyres south of the ACC are evident in the Ross and Weddell Seas.

Growing evidence from observations and models suggests that the frontal structure of the ACC is richer than suggested by earlier studies. The 3-year mean surface velocity field in SOSE shows that the ACC consists of multiple narrow filaments (Figure 18.1b). The jet structure is even more complicated in a 5-day snapshot (Figure 18.1c), as expected, but it is notable how much of this structure remains in the 3-year mean field. The multiple jets are a robust characteristic of the ACC, both at the surface and at depth (Figure 18.1d). Multiple ACC jets have now been observed in a number of data sets, including repeat hydrographic sections (Sokolov and Rintoul, 2007), repeat acoustic Doppler current profiler (ADCP) transects (Firing et al., 2011), sea surface temperature data

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(Hughes and Ash, 2001), dynamic height fields derived from surface drifters (Niiler et al., 2003; Hughes, 2005), and sea surface height measurements (Sokolov and Rintoul, 2007, 2009a,b; Hughes et al., 2010). The current branches are observed to merge and diverge along the circumpolar path and to vary in time (see also Chapter 12). Numerical simulations with sufficient spatial resolution produce a filamented ACC similar to observations (e.g., Hallberg and Gnanadesikan, 2006; Mazloff et al., 2010; Thompson et al., 2010). Recognition of the existence of multiple jets in the ACC has provided a new and useful perspective from which to view the eddy–mean flow and flow–topography interactions that are central to ACC dynamics (e.g., Hughes, 2005; Thompson et al., 2010; Thompson and Sallée, 2012). The jets result from eddy-mean flow interactions and, in turn, the structure of the jets determines the stability of the flow and the nature of the eddy field. Interaction of the jets with topography influences eddy generation and mixing and establishes the bottom stresses and pressure torques that enable closure of the momentum and vorticity budgets, respectively.

3.2. Transport of the ACC

The ACC is the primary means of interbasin exchange and substantial effort has been devoted to measuring the ACC transport. Observing the mean absolute transport of the ACC remains a difficult challenge. The geostrophic transport relative to a level of no motion (the baroclinic transport) can be determined in a straightforward way from measurements of temperature and salinity (although few sections are repeated with sufficiently high frequency to avoid aliasing unresolved variability). The velocity at the level of no motion (often taken to lie near the sea floor in the Southern Ocean), and hence the barotropic transport, needs to be determined by long-duration, closely spaced direct velocity measurements (e.g., current meter moorings) or inferred indirectly (e.g., by inverse methods). The difficulty of directly observing the barotropic transport over large space and timescales means that studies have often focused on the baroclinic contribution to the total transport. When discussing variations in transport, it can be important to distinguish between net interbasin exchange (i.e., the total flow between Antarctic and the southern hemisphere continents) and the transport of the ACC itself, as noted by Hughes et al. (1999). The former may be sensitive to phenomena on the boundaries (e.g., the southern mode, discussed below, or boundary waves) and to extensions of the subtropical current systems crossing the northern end of sections spanning the Southern Ocean (e.g., south of Africa and Australia).

Meredith et al. (2011) provide a summary of historical and recent measurements of the ACC transport in Drake Passage, the best-measured region of the Southern Ocean. The mean baroclinic volume transport for 16 sections occupied between 1993 and 2009 was 136.7 Sv, with a standard deviation of 6.9 Sv, and no indication of a trend. Four years of inverted echo sounder measurements spanning Drake Passage gave a mean full-depth baroclinic transport of 126.7 ± 1.5 Sv, with root mean square (rms) fluctuations of 9.5 Sv (Chidichimo et al., 2013). The mean absolute transport above 1000 m depth in Drake Passage was 95 Sv, with a standard error of 2 Sv, based on the average of 53 repeat ADCP sections between 2005 and 2009 (Firing et al., 2011). The mean transport through Drake Passage in SOSE, illustrated in Figure 18.1, is 153 ± 5 Sv (Mazloff et al., 2010). Measurements of the ACC transport at other locations (e.g., south of Africa (Swart et al., 2008) and south of Australia (Rintoul et al., 2002)) are largely consistent with those made at Drake Passage, once the additional transport south of Australia needed to balance the Indonesian Throughflow is taken into account.

As found in the studies cited earlier, estimates of the ACC transport from observations and models consistently show that the variability in ACC transport on timescales of years to decades is small relative to the transport's mean value (e.g., Rintoul et al., 2002; Cunningham et al., 2003; Meredith et al., 2004; Hallberg and Gnanadesikan, 2006; Böning et al., 2008). The relative stability of the ACC transport has motivated new efforts aimed at understanding the response of the ACC to climatic perturbations in wind and buoyancy forcing.

3.3. Response of the ACC to Wind and Buoyancy Forcing

Both wind and buoyancy forcing can alter the slope of isopycnals across the Southern Ocean, and therefore the transport of the ACC. Variability and trends in wind and buoyancy forcing may arise from internal variability of the climate system, including teleconnections to low-latitude phenomena such as the El Niño–Southern Oscillation, and from anthropogenic drivers including global atmospheric warming and loss of stratospheric ozone. The response of the ACC depends on the frequency of the forcing, with high-frequency forcing favoring a barotropic response and baroclinic mechanisms becoming more important at lower frequencies.

3.3.1. Response to High-Frequency Wind Forcing: The Southern Mode

For wind-forcing perturbations on timescales shorter than O (1 year), the variability in Antarctic circumpolar transport is dominated by a barotropic mode that follows f/H contours (where f is the Coriolis parameter and H is the ocean depth), and which is therefore focused along the southern edge of the Southern Ocean. This was termed the "southern mode" by

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Hughes et al. (1999). Propagation of wind-forced barotropic topographic Rossby waves along *f*/*H* contours governs the dynamics of this mode. The characteristic propagation speed of the waves is several meters per second, yielding a circumpolar adjustment timescale of a few days (Webb and de Cuevas, 2007). The mode has been shown to explain a substantial proportion of the daily-to-interannual variance in sea level and bottom pressure near the Antarctic continent, and in the transport through Drake Passage, in both observations and eddy-permitting general circulation models (Aoki, 2002; Hughes et al., 2003; Zika et al., 2013a).

3.3.2. Response to Low-Frequency Wind Forcing: Eddy Saturation

On timescales longer than O(1 year), the response of the ACC to forcing perturbations is distinct in both spatial structure and dynamical character compared with the shorter timescale response. In mid-latitudes, variations in wind forcing excite baroclinic Rossby waves that propagate westward and are instrumental in the adjustment of the basin-scale gyres to forcing perturbations, confining the response to the upper ocean, and establishing western boundary currents (Anderson and Killworth, 1977; Hughes and de Cuevas, 2001). The dynamical balance of the ACC is distinct from that of mid-latitude basins in two subtle yet profound aspects. First, the eastward flow of the ACC exceeds the westward phase speed of baroclinic Rossby waves (i.e., the flow is supercritical) and baroclinic disturbances propagate eastward in the jets of the ACC (Hughes, 1996). As baroclinic waves are unable to propagate westward in the ACC, the response to wind forcing is not limited to the upper ocean there and the ACC extends to greater depth than the mid-latitude gyres. Second, owing to the absence of continental boundaries at the latitude band of Drake Passage, variations in wind forcing must be balanced by topographic form drag on submarine obstacles in the path of the ACC, rather than pressure gradients along the boundaries of the basin (e.g., Hughes and de Cuevas, 2001). Therefore, in order for the ACC to reach a stationary state, the horizontal momentum imparted by wind forcing must be transported vertically to great depth before it can be removed by submarine topography. This interfacial form stress is effected by mesoscale eddies (e.g., Olbers et al., 2004). In the following, we will argue that these two distinctive features of the dynamics of the ACC and the details of their interaction are key to understanding the response of the current to perturbations in forcing.

Models and observations suggest that the ACC responds to an increase in wind forcing primarily by increasing eddy kinetic energy (EKE) rather than increasing transport, for realistically strong wind forcing (e.g., Tansley and Marshall, 2001; Cunningham et al., 2003; Hallberg and Gnanadesikan, 2006; Meredith and Hogg, 2006; Böning et al., 2008; Firing et al., 2011; Munday et al., 2013). The limit in which the time-mean ACC transport is insensitive to changes in wind forcing, first discussed by Straub (1993), has come to be known as the "eddy saturation" regime. The models indicate that stronger wind forcing increases the baroclinicity of the ACC; baroclinic instability then releases available potential energy (reduces isopycnal tilt) and increases EKE, reestablishing the pre-perturbation baroclinic state. Eddy saturation therefore describes the stationary-state response to the change in forcing, rather than implying completely invariant ACC transport. The characteristic time lag between the wind perturbation and the change in EKE is typically a few years in models. An analysis of altimeter data suggested a similar delay in the EKE response to a strong wind perturbation in 1998 (Meredith and Hogg, 2006), while a subsequent investigation of a longer period of altimetric measurements found that the EKE response varied strongly with location and with the nature of the wind perturbation (Morrow et al., 2010). Models and observations indicate that the ACC approaches, but does not reach, the eddy saturation limit, with most models showing a transport increase of less than 10% to a doubling of the wind stress (e.g., Hallberg and Gnanadesikan, 2006; Farneti et al., 2010; Dufour et al., 2012).

A complete theory of eddy saturation does not yet exist. Straub (1993) used a simple theoretical argument to show that once the ACC reached the threshold for baroclinic instability, the transport of the current would be independent of the wind forcing. However, the details of how interactions between the mean flow, the standing and transient eddy fields, and topography establish the eddy-saturated response to changes in forcing are only beginning to emerge. A common thread is the significance of local dynamics. The standing and transient eddy fields, and hence the eddy fluxes of momentum and vorticity, vary along the path of the current, and these variations are likely to be important to the dynamics of the ACC as a whole. For example, a transient strengthening of the ACC in response to an increase in wind forcing might alter the standing meander pattern and hence the vertical momentum transfer by standing and transient eddies (e.g., Dufour et al., 2012; Thompson and Naveira Garabato, 2013). The change in interfacial form stress can then accelerate the deep flow of the ACC and produce additional topographic form drag to balance the wind forcing and restore the pre-perturbation conditions. The response of the ACC to changes in wind forcing also depends on the spatial structure of the wind perturbation (e.g., Morrow et al., 2010; Mazloff, 2012), again underscoring the importance of local dynamics. Yet another example is the likely importance of internal waves radiated by small-lateral-scale (O(1–10 km)) topography in arresting the ACC flow (e.g., Naveira Garabato et al., 2013a) and in dissipating the Southern Ocean mesoscale eddy field (e.g., Nikurashin et al., 2013; Waterman et al., 2013).

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3.3.3. Response to Buoyancy and Remote Forcing

Any local or remote forcing that alters the meridional density gradient across the ACC can change the transport. In addition to the wind effects discussed above, buoyancy forcing and processes that influence the depth of the pycnocline on the northern flank of the Southern Ocean can also shape the baroclinicity of the ACC. The sensitivity of ACC transport to buoyancy forcing has been discussed by a number of authors (e.g., Gent et al., 2001; Borowski et al., 2002; Hogg, 2010). Cooling in the south and warming in the north will tend to increase the tilt of isopycnals and therefore the baroclinic transport. Buoyancy forcing may also alter potential vorticity (PV) gradients in the upper ocean and therefore the "slumping" of isopycnals by eddy PV fluxes (e.g., Marshall and Radko, 2003).

The baroclinicity of the ACC is closely related to the depth and structure of the global pycnocline. Processes that affect the meridional slope of isopycnals in the Southern Ocean influence the global pycnocline (e.g., Wolfe and Cessi, 2010; Chapter 11), and processes outside the Southern Ocean that affect the global pycnocline influence the meridional slope of isopycnals in the ACC. The latter set of processes includes deep water formation in the North Atlantic (e.g., Gnanadesikan 1999; Hallberg and Gnanadesikan, 2001), interior diapycnal mixing at mid- and low latitudes (e.g., Nikurashin and Vallis, 2011), and wind forcing to the north of the ACC (Allison et al., 2011). Anomalies in mid- and low-latitude pycnocline depth induced by forcing both local to and remote from the ACC trigger a global wave adjustment lasting between many decades and several centuries, where the long timescale relates to the large volume of water resident in the global pycnocline that must be displaced vertically for the adjustment to occur (e.g., Allison et al., 2011; Jones et al., 2011). Evidence of the ACC's baroclinic adjustment to low-latitude wind forcing is provided by Close and Naveira Garabato (2012), who show that isopycnals in the upper kilometer of northern Drake Passage have deepened markedly between the early 1990s and 2010 in the wake of poleward-propagating boundary waves generated by changes in tropical Pacific winds. The baroclinic adjustment of the ACC therefore occurs on multiple timescales: an interannual response to local forcing, and a multidecadal-to-centennial response set by the timescale for equilibration of the ACC's baroclinic structure with the stratification of the ocean basins to the north. The existence of this longer timescale provides an alternative explanation for the observed resilience of the ACC transport, though wind perturbation experiments with idealized eddy-resolving models run to equilibrium (Munday et al., 2013) suggest that eddy saturation ideas may apply on these long timescales as well.

4. SOUTHERN OCEAN OVERTURNING CIRCULATION

The existence of a double-celled overturning circulation in the Southern Ocean (Figure 18.2) was inferred in the early days of oceanography from observed hydrographic distributions. However, the structure and transport of the overturning have been difficult to quantify. In this section, we first discuss the water mass transformations at the heart of the overturning circulation and review recent progress toward a quantitative representation of the overturning. We then provide an overview of the insights gained from residual-mean theory and introduce the concept of (partial) eddy compensation. Eddy stirring of PV provides a useful framework to consider interactions between eddies, the mean flow, and topography and exposes potential limitations of the zonally averaged view of the Southern Ocean circulation.

4.1. Water Mass Transformations and Southern Ocean Overturning

The role of the Southern Ocean in the global overturning circulation can be appreciated by following a water parcel along the global circuit, starting in the North Atlantic. North Atlantic Deep Water (NADW) flows southward in the Atlantic, largely along isopycnals, to reach the Southern Ocean. Some of the NADW flows poleward across the Southern Ocean, where it forms a high-salinity layer known as Lower Circumpolar Deep Water (LCDW). The LCDW outcrops near Antarctica, where cooling and brine rejected during sea ice formation transform LCDW to denser AABW. The poleward spreading of LCDW and equatorward return flow of AABW form the lower cell of the Southern Ocean overturning circulation (Figure 18.2).

LCDW also enters the deep Indian and Pacific Oceans, where enhanced diapycnal mixing at depth converts the inflowing water to slightly less dense Upper Circumpolar Deep Water (UCDW), which is returned to the Southern Ocean (e.g., Lumpkin and Speer, 2007). The UCDW is associated with an oxygen minimum layer, reflecting the relatively long isolation from the atmosphere during its transit of the deep ocean basins. UCDW spreads poleward, above the LCDW, and outcrops within the ACC belt. Warming and freshening of the surface layer converts UCDW into less dense Antarctic Intermediate Water (AAIW) and Subantarctic Mode Water (SAMW), which flow equatorward to ventilate the pycnocline of the Southern Hemisphere oceans. The poleward flow of UCDW and equatorward flow of AAIW and SAMW form the upper cell of the Southern Ocean overturning circulation (Figure 18.2). A third cell, sometimes called the subtropical cell, involves poleward transport of subtropical waters that are transformed to cooler, denser SAMW before returning equatorward, and will not be discussed here (see Chapter 9).

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The classical view of the global overturning circulation emphasized a diffusive return path: sinking of dense water in the polar regions was balanced by uniform upwelling and vertical mixing in the ocean interior (e.g., Munk, 1966). Direct measurements later showed the rate of interior mixing to be up to an order of magnitude too small to support the rate of upwelling required for mass balance (e.g., Ledwell et al., 1993; Chapter 7). These observations helped motivate studies of overturning in the weak mixing limit (e.g., Toggweiler and Samuels, 1998). In this limit, water mass transformations driven by buoyancy forcing in the Southern Ocean, rather than diapycnal mixing in the ocean interior, act to connect the upper and lower limbs of the global overturning circulation, while the interior flow is largely adiabatic.

Recent observations, theory and numerical simulations suggest that both adiabatic and diabatic processes are important. While the overturning circulation is largely adiabatic in the ocean interior, diapycnal mixing drives water mass transformations that are central to the global overturning. The conversion of LCDW to UCDW in the deep Indian and Pacific oceans is one example. Efforts to quantify the water mass transformations driven by breaking of internal waves are just beginning. An illustration is provided by Figure 18.3, where estimates of the energy input to internal waves by tides and geostrophic motions and a turbulent mixing parameterization are used to quantify the diapycnal water mass transformations driven by diapycnal mixing (Nikurashin and Ferrari, 2013). Mixing is largest in the deep ocean and decays with height above the bottom. While such estimates are uncertain due to numerous assumptions and poorly constrained parameters, they suggest that mixing by breaking internal waves makes a significant contribution to water mass transformation in the abyssal ocean. Observations show that diapycnal mixing is enhanced in the deep Southern Ocean where the ACC jets interact with rough bathymetry (e.g., Naveira Garabato et al., 2004; Waterman et al., 2013).

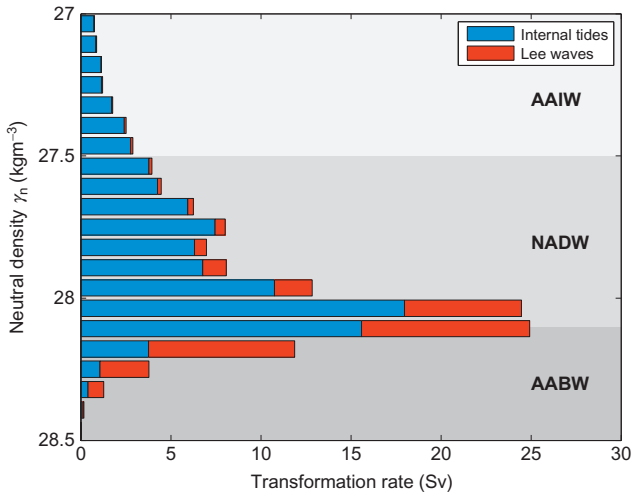

FIGURE 18.3 The water mass transformation rate (Sv) driven by breaking of internal waves, as a function of neutral density, with the density range of different water masses indicated. Both internal tides (blue) and deep geostrophic flows (red) contribute to generation of internal waves and diapycnal mixing; the latter is particularly important in the ACC. While there is growing evidence to support the shape of the curve, with elevated mixing at depth and weak mixing in the upper ocean, the magnitude of the implied water mass transformations is sensitive to a number of assumptions and poorly known parameters. The density range of Antarctic Bottom Water (AABW), North Atlantic Deep Water (NADW), and Antarctic Intermediate Water (AAIW) are shown. *From Nikurashin and Ferrari (2013).*

Diapycnal mixing in the thermocline and upper ocean is generally weak, even in the Southern Ocean (e.g., Naveira Garabato et al., 2004; Nikurashin and Ferrari, 2013). The conversion of UCDW to intermediate waters required to close the global overturning circulation is instead accomplished by air–sea buoyancy fluxes and diapycnal mixing by horizontal eddy motions in the surface mixed layer (Speer et al., 2000; Marshall and Speer, 2012).

In summary, the Southern Ocean plays several key roles in the global overturning circulation. The circumpolar flow of the ACC provides the interbasin connection that permits a global-scale overturning to exist. The steeply sloping isopycnals of the ACC allow upwelling of deep water to the sea surface, where air–sea buoyancy forcing and enhanced diapycnal mixing in the surface layer act to transform deep water to intermediate waters, thereby connecting the deep and shallow limbs of the global overturning. Intermediate waters (AAIW + SAMW) exported from the Southern Ocean ultimately supply the northward transport in the upper layers of the Atlantic required to balance the export of NADW (e.g., Sloyan and Rintoul, 2001; Marshall and Speer, 2012). Note, however, that NADW/LCDW is too dense to outcrop in regions of net ocean buoyancy gain in the ACC belt, as is often assumed in conceptual models of the zonally averaged overturning circulation. LCDW must first be transformed to less dense UCDW by diapycnal mixing in the deep Indian and Pacific oceans. Vigorous diapycnal mixing by internal wave breaking in the deep Southern Ocean helps close the lower cell of the overturning by converting AABW to CDW. The Southern Ocean overturning therefore lies somewhere between the adiabatic and diabatic limits: while much of the flow associated with the overturning cells is adiabatic, and mixing is weak in the pycnocline, water mass transformations driven by diapycnal mixing in the deep ocean, by air–sea buoyancy exchange, and by diapycnal mixing in the surface mixed layer are fundamental to the overturning.

4.2. Estimates of the Rate of Southern Ocean Overturning

Quantifying the strength of the Southern Ocean overturning circulation has long been a challenge. Small biases in the observed or inferred absolute horizontal velocity can accumulate to large errors on long sections, while vertical motions are not observable with present technology. In

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addition, the meridional flow associated with the overturning is a small residual of larger wind- and eddy-induced transports and it depends on a variety of small- and mesoscale processes that are difficult to measure (e.g., turbulent diapycnal mixing and eddy stirring). Nonetheless, some progress has been made in quantifying the net overturning by combining observations and dynamical constraints with a variety of inverse methods. Recent estimates suggest that the strength of the upper cell across 32°S is about 12–21 Sv (with largely overlapping uncertainties; Ganachaud, 2003; Lumpkin and Speer, 2007; Mazloff et al., 2010; Naveira Garabato et al., 2013b), consistent with estimates of 15 Sv based on calculations of water mass transformations driven by climatological surface forcing (e.g., Olbers et al., 2004).

There is less agreement on the partitioning of the zonally averaged upper-cell flow among the three major ocean basins. While most inverse studies find about 20 Sv of water in the CDW density class enter the Southern Ocean from the subtropical Atlantic (e.g., Sloyan and Rintoul, 2001; Ganachaud, 2003; Lumpkin and Speer, 2007; Mazloff et al., 2010; Naveira Garabato et al., 2013b), there are substantial discrepancies in the Indian and Pacific oceans. Most recent studies estimate that about 10 Sv of deep water flows poleward in each of these two basins, but the distribution of the flow with respect to density differs, affecting the degree of cancellation in the zonal integral. Finally, investigations of the vertical circulation on subbasin scales suggest that zonally averaged estimates conceal important spatial variations in the strength and sense of overturning. For example, the subduction of fluid across the base of the winter mixed layer to supply the SAMW and AAIW layers occurs in highly localized regions, as set by the combined influence of wind, eddies, and lateral induction resulting from along-stream variations in winter mixed layer depth (Figure 18.4; Sallée et al., 2010a; Herraiz-Borreguero and Rintoul, 2011).

In contrast to studies of the upper cell, estimates of the strength of the lower cell still vary over a wide range. Inverse studies explicitly representing the circulation of the subpolar Southern Ocean (e.g., Lumpkin and Speer, 2007; Mazloff et al., 2010) tend to estimate a relatively weak lower cell of ca. 10 Sv across nominally 60°S, in reasonably good alignment with transient tracer-based estimates of AABW ventilation (Orsi et al., 2002). Naveira Garabato et al. (2013b) estimate a somewhat larger 15 Sv. Entrainment tends to increase the transport of deep and bottom waters flowing northward in the lower cell and the deep overturning circulation tends to be stronger near 30°S than at 60°S. Recent estimates of the deep overturning at 30°S (e.g., 18.6 ± 0.9 Sv (Naveira Garabato et al., 2013b), 20.9 ± 6.7 Sv (Lumpkin & Speer, 2007), 29 Sv (Talley, 2013)) tend to be somewhat smaller than the O(50 Sv) estimated earlier by Schmitz (1996) and Sloyan and Rintoul (2001). Part of the difficulty in quantifying the strength of the lower cell arises from the fact that the northward and southward limbs of the cell are connected by poorly known diapycnal mixing processes in the deep ocean. Direct measurements of the equatorward export of AABW at selected outflow sites have helped quantify the pathways of the lower cell. For example, northward AABW transports of ca. 6 Sv through the Orkney Passage (north of the Weddell Sea) and ca. 12 Sv in the deep boundary current east of the Kerguelen Plateau have been measured by Naveira Garabato et al. (2002) and Fukamachi et al. (2010), respectively. However, comparison of transport estimates near the sources of AABW with estimates in the export pathways is not straightforward, as entrainment and recirculation may increase the transport downstream of the sources.

4.3. Residual-Mean Circulation

Despite the intrinsically three-dimensional nature of the Southern Ocean overturning circulation, much theoretical progress has been made by considering a zonally averaged view of the circulation. Residual-mean theory considers the time-mean, steady, zonal- (or streamwise-) average buoyancy and zonal momentum budgets of the ACC to assess how the interplay between wind and surface buoyancy forcing, diapycnal mixing, and mesoscale ocean dynamics shapes both the zonal transport and the Lagrangian-mean overturning across the region (Marshall and Radko, 2003; Olbers et al., 2004 and references therein). It is in making explicit the dynamical connection between zonal and meridional (overturning) flow that the residual-mean approach is particularly insightful. In short, meridional flow is induced by the vertical divergence in a zonal stress, which is initially imparted on the ocean by the wind, transmitted downward by transient eddies (as an interfacial form stress), and ultimately opposed by topographic form drag at the sea floor.

An equivalent description of the residual-mean overturning circulation is as the sum of the wind-driven and eddy-induced circulations. This description is illustrated in Figure 18.5, which presents results from an eddy-resolving channel model of the circumpolar current (Abernathey et al., 2011; Marshall and Speer, 2012). The wind-driven (Eulerian mean) overturning cell consists of northward flow in the surface Ekman layer, a southward return flow at depth, and upwelling (downwelling) to the south (north) of the current's axis. It thus acts to tilt isopycnals. Release of the potential energy stored in the sloping isopycnals by baroclinic instability generates transient eddies. The eddy-induced overturning opposes the Ekman cell, driving a slumping of isopycnals, a tendency referred to as "eddy compensation". The sum of the wind-driven and eddy-induced circulations yields the residual-mean circulation, which is responsible for tracer transports as illustrated by the simulated "dye release" in Figure 18.5.

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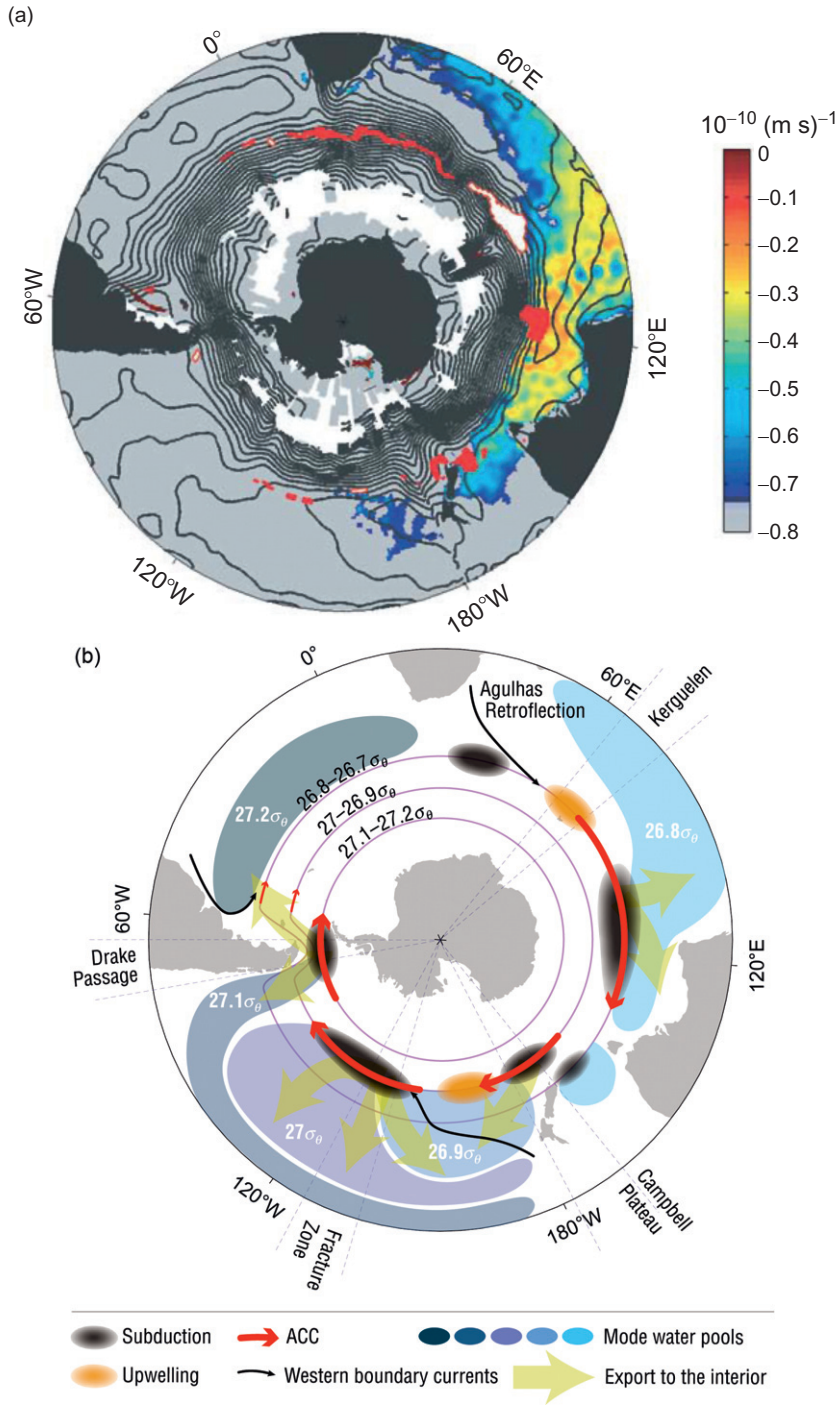

FIGURE 18.4 Subduction hot spots in the Southern Ocean inferred from Argo floats, satellite altimetry, and wind forcing (see Sallée et al., 2010a for details). (a) Subduction on the $26.8\sigma_\theta$ surface (red dots), overlaid on the Montgomery streamlines indicating the flow on this density surface (black lines). The potential vorticity on the isopycnal is shown in color. A tongue of low-potential vorticity spreads along streamlines from the subduction hot spot in the southeast Indian Ocean into the interior (anthropogenic CO_2 follows similar pathways; Sallée et al., 2012). (b) A schematic view of subduction in the Southern Ocean. Each density surface is ventilated through a narrow window, with denser surfaces ventilated progressively further east from the Indian Ocean. Subduction in some locations is ineffective, because subducted water is upwelled again further downstream (e.g., in the central Indian and western Pacific basins, regions of net upwelling shown in orange). Subduction and upwelling are defined as transfer across the base of the winter mixed layer. *From Sallée et al. (2010a).*

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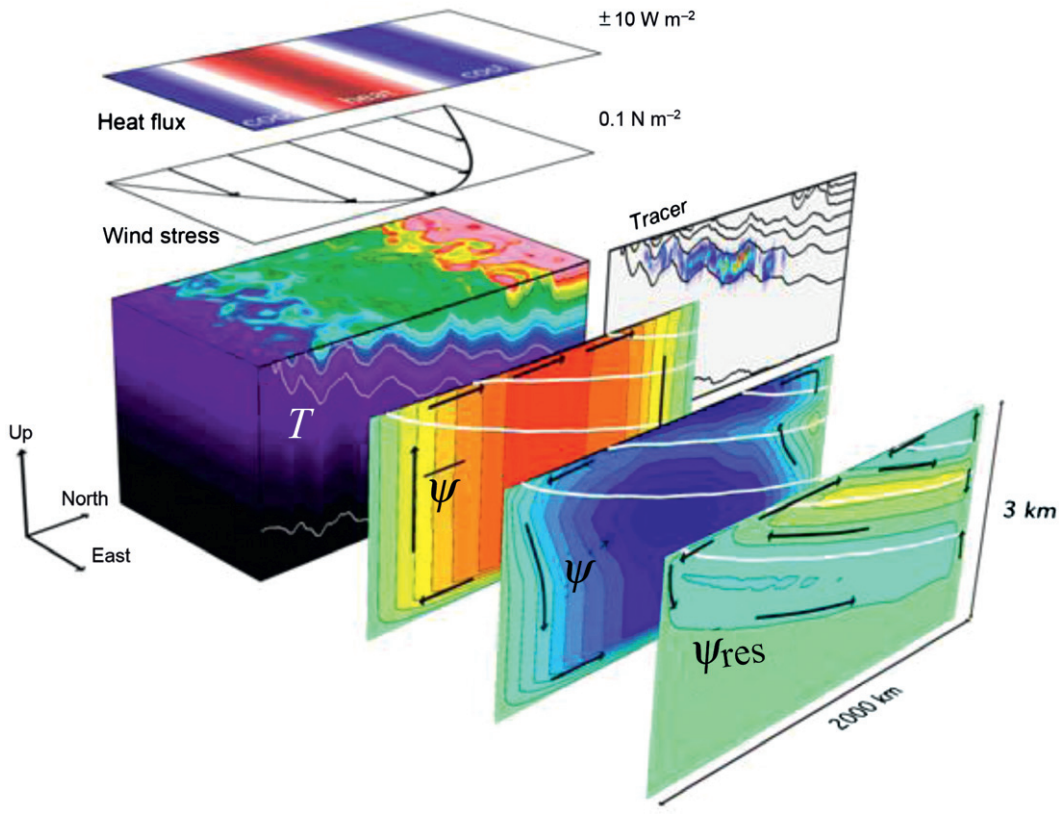

FIGURE 18.5 An illustration of the residual-mean circulation in the Southern Ocean, derived from an idealized but eddy-resolving ocean circulation model (Abernathey et al., 2011; see http://vimeo.com/17822148 for a movie of the spin-up of the circulation). The reentrant channel is forced by wind and heat fluxes at the sea surface. From left to right, the lower panels show a snapshot of the temperature (T) field; the time-mean overturning associated with the wind-driven Ekman transport ($\bar{\Psi}$); the time-mean counter-rotating overturning associated with eddy fluxes (Ψ^*); and their sum, the residual-mean overturning streamfunction (Ψ_{res}). The panel labeled "Tracer" illustrates the poleward, along-isopycnal transport of tracer by the residual-mean flow. *From Marshall and Speer (2012), redrawn from Abernathey et al. (2011).*

In the residual-mean framework, the strength of the net overturning is set by the buoyancy forcing, which transforms upwelled deep water to less dense intermediate water. Both air–sea exchange and diapycnal mixing in the surface mixed layer can contribute to the net buoyancy forcing.

The tendency for the eddy-induced circulation to compensate the wind-driven circulation is a common feature of models that resolve or effectively parameterize the effect of eddies. (Coarse resolution models tend to underestimate the eddy-driven circulation and therefore overestimate the sensitivity of the overturning to changes in wind stress (e.g., Farneti et al., 2010).) For example, increasing the wind stress in a reentrant sector model of the Southern Ocean results in an increase in EKE, a small increase in ACC transport, and a larger increase in overturning (Figure 18.6; Morrison and Hogg, 2013). However, the degree of compensation varies widely between models. For a doubling of wind stress, ACC transport increases by about 25% and overturning increases by about 50–70% in Morrison and Hogg's (2013) model. Many other models are closer to the eddy saturation and eddy compensation limits, showing an increase in ACC transport of less than 10% and an increase in overturning of less than 25% for a doubling of wind stress (e.g., Hallberg and Gnanadesikan, 2006; Farneti et al., 2010; Abernathey et al., 2011; Dufour et al. 2012). Detailed comparisons are difficult to make because the model simulations differ in many respects, including geographic domain, model resolution and complexity, and the nature of the imposed wind-forcing perturbation. Perhaps most importantly, the effective buoyancy forcing applied is likely to have a strong impact on the sensitivity of the overturning response. We note that the coupled model study of Farneti et al. (2010), in which the buoyancy flux adjusts to changes in the ocean and atmosphere, is the closest to the eddy saturation and eddy compensation limits.

Despite quantitative disagreement between results from different models, dynamical arguments (Meredith et al., 2012) and the modeling studies cited above provide a consistent picture of the response of the Southern Ocean circulation to changes in wind forcing. Both the ACC transport and overturning circulation are less sensitive to wind-forcing perturbations than would be expected in a non-eddying ocean, and the ACC transport is less sensitive than the overturning circulation to wind changes. While the

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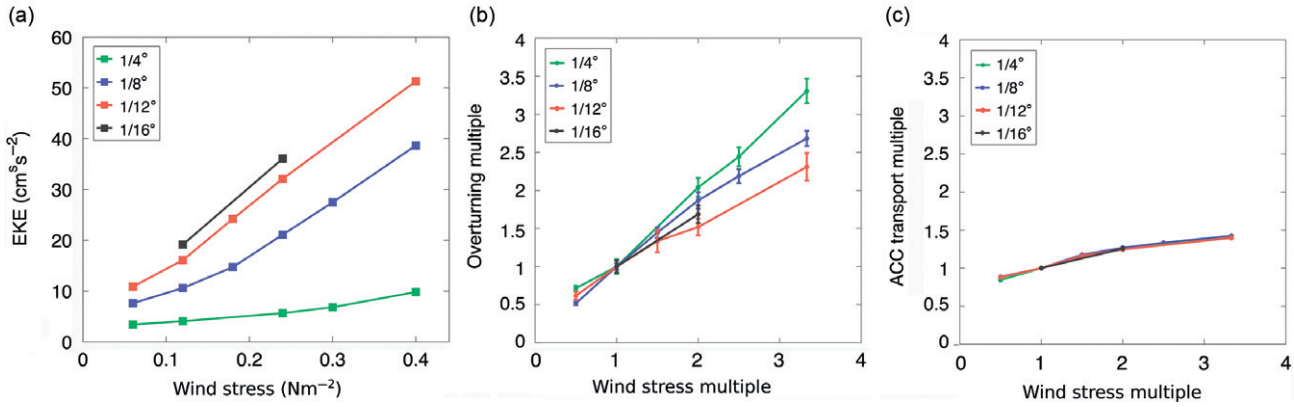

FIGURE 18.6 Eddy compensation and eddy saturation. Integrations of an idealized ocean circulation model illustrate the concepts of eddy saturation and eddy compensation, and their dependence on the extent to which eddies are resolved. (a) As wind forcing increases, eddy kinetic energy (EKE) increases, with a tendency for greater increase in EKE with increasing model resolution. (b) Sensitivity of the overturning circulation at 30°S to wind forcing (transport and wind stress have been scaled relative to a control case). The overturning strength increases with wind stress, but more slowly in models that better resolve eddies, illustrating that eddy-driven overturning partially compensates wind-driven changes. Zero eddy compensation would follow a diagonal (1:1) line; perfect eddy compensation would correspond to a horizontal line. (c) Sensitivity of ACC transport to wind forcing. A linear increase with wind would follow a diagonal (1:1) line; perfect eddy saturation would fall on a horizontal line. *Redrawn from Morrison and Hogg (2013), figure supplied by A. Morrison.*

degree of eddy saturation and eddy compensation tends to increase when refining model resolution, the simulations do not reach the eddy saturation or eddy compensation limits, even with very fine resolution. Unless surface buoyancy forcing is fixed, neither the magnitude nor the shape of the vertical profile of eddy-induced stress divergence will necessarily match that of the Ekman component, resulting in incomplete compensation.

4.4. Eddy Stirring of PV

While model results suggest that eddies tend to partially compensate the wind-driven circulation response to a change in wind forcing, the amount of compensation remains uncertain. Progress requires a better understanding of the eddy-induced circulation. This circulation arises from transient eddy fluctuations on scales of O(10–100 km) that are difficult to quantify in observations and to describe theoretically. A common approach is to relate the stress divergence (or eddy-induced meridional flow) to a downgradient flux of PV resulting from eddy stirring of a background isopycnal PV gradient, where the efficiency of the eddy stirring process is characterized by an isopycnal diffusivity (e.g., Rhines and Young, 1982). This approach reduces the overturning problem to an investigation of the physical controls of the isopycnal diffusivity and its spatial covariability with the background isopycnal PV gradient. The simplification is artificial in that the background PV distribution is itself shaped by the action of eddies. However, the substantial resilience of the mean PV gradient to changes in forcing in an eddy-saturated regime (Meredith et al., 2012) suggests that much may be learned about the dynamical controls of the Southern Ocean overturning by focusing on the physics of eddy stirring.

Recognition of the importance of eddy stirring in the Southern Ocean overturning circulation has motivated a recent flurry of investigations on the subject (e.g., Marshall et al., 2006; Abernathey et al., 2010; Ferrari and Nikurashin, 2010; Naveira Garabato et al., 2011; Thompson and Sallée, 2012). Two general findings have emerged from that body of work. First, the intensity of eddy stirring is not solely governed by EKE, contrary to the predictions of classical quasi-homogeneous turbulence theories. Rather, it is commonly reduced at the upper-ocean core of the ACC frontal jets (where EKE is often highest) relative to surrounding regions, in conjunction with large isopycnal PV gradients (e.g., Thompson et al., 2010; Naveira Garabato et al., 2011). Second, jets are found to be leaky or permeable in the proximity of prominent topographic obstacles, where the mean flow adopts a nonzonal configuration (Figure 18.7). There, vigorous eddy stirring may occur in association with large isopycnal PV gradients, suggesting that those sites are major contributors to the circumpolar integral of the cross-stream PV flux, vertical divergence in along-stream stress, and ultimately cross-stream overturning, as found by, for example, Zika et al. (2013b) in an eddy-permitting GCM. The localization of eddy fluxes by topography provides another example of how the zonally averaged view might conceal important aspects of Southern Ocean dynamics.

The reduction of isopycnal diffusivity in the upper layers of the ACC frontal jets may be interpreted in at least two alternative, partially compatible ways. Some authors (e.g., Marshall et al., 2006; Abernathey et al., 2010), treating mesoscale eddies as linear Rossby waves, suggest that the intensification of eddy stirring on the flanks of jets is associated with a Rossby wave critical layer, as in the atmospheric literature on eddy stirring. Others (Ferrari and Nikurashin, 2010; Naveira Garabato et al., 2011) note

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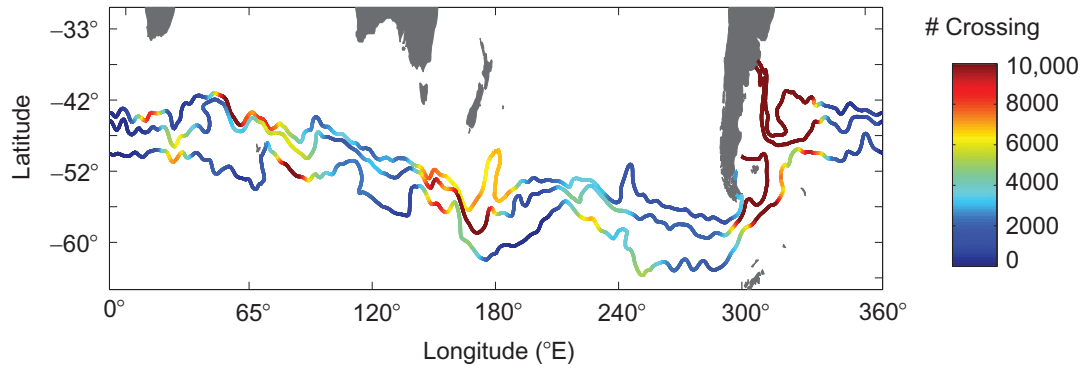

FIGURE 18.7 Number of particle crossings (a metric of the rate of eddy stirring) across (from north to south) the northern branch of the Subantarctic Front, the main branch of the Subantarctic Front, and the Polar Front, from a synthetic release of Lagrangian particles in the altimetric velocity field by Thompson and Sallée (2012). Note the localization of particle crossings to nonzonal segments of the fronts, which correspond to regions where the ACC fronts are steered by topographic obstacles. *From Thompson and Sallée (2012).*

that oceanic mesoscale eddies possess a significant degree of nonlinearity, which results in kinematic suppression of the eddy mixing length (the typical cross-stream displacement of water parcels induced by stirring) by eddy–mean flow interaction. By contrast, there is as yet no consensus on how to interpret the permeability of ACC jets in nonzonal segments of the current close to topography. Using ray-tracing arguments and evidence from idealized studies (e.g., Thompson, 2010), Naveira Garabato et al. (2011) conjecture that nonparallel structure in the mean flow, such as that associated with standing meanders, leads to a breakdown of the conditions required for suppression of eddy stirring: eddies are locally strained, amplified, and eventually "torn" by narrow and twisted jets rather than advected as coherent entities by broader, parallel jets.

The preceding discussion suggests that in the limit of an invariant PV distribution, both the wind-driven Ekman and eddy-induced components of the overturning are expected to respond positively to wind-forcing perturbations. The eddy-induced overturning increases because isopycnal eddy diffusivity depends on EKE, and hence on wind stress (Meredith et al., 2012). As the vertical profiles of the stress divergence associated with the Ekman and eddy contributions to the overturning are generally different, eddy compensation will not be complete and the net (residual) Southern Ocean overturning will scale with changes in wind forcing.

Nonetheless, the extent to which the limit of invariable PV applies in the Southern Ocean is uncertain. The background PV field can be modified by wind (e.g., cross-frontal advection in the Ekman layer) or buoyancy forcing (e.g., convection) (Thomas, 2005), or by eddy buoyancy fluxes in the surface mixed layer (Marshall and Radko, 2003). Any modification of the upper-ocean PV field upon which eddy stirring acts will change the eddy flux of PV along isopycnals, the vertical profile of stress divergence, and the eddy-induced overturning. Various mechanisms can alter the PV distribution in the interior as well, including the wave-mediated adjustment of low-latitude stratification to local and remote forcing (e.g., Allison et al., 2011; Munday et al., 2013) and the modification of stratification by changes in diapycnal mixing (e.g., Ito and Marshall, 2008; Nikurashin and Vallis, 2011). Diapycnal mixing in the deep Southern Ocean may depend on wind forcing, either through wind-forced barotropic fluctuations in the intensity of deep boundary currents (Polzin and Naveira Garabato, 2013) or by changes in internal wave generation and breaking produced by wind-driven changes in the ACC eddy field interacting with rough topography (Nikurashin et al., 2013; Waterman et al., 2013).

5. SOUTHERN OCEAN CHANGE

Over the last decade, evidence has grown of strong variability and trends in water properties in the Southern Ocean. Given the influence of Southern Ocean processes on the global ocean circulation, climate, and biogeochemical cycles, change in the circulation of the Southern Ocean might have widespread consequences. Observations of the response to changes in natural and anthropogenic forcing can also provide useful insights into the dynamics of the system. Here we briefly summarize recent evidence of changes in the Southern Ocean. The observational record is short and in most cases insufficient to distinguish long-term trends from low-frequency variability.

5.1. Warming and Freshening of the Southern Ocean

The upper 2000 m of the Southern Ocean has warmed at a faster rate and to greater depth than the global ocean average in recent decades. Gille (2008) demonstrated that the upper 1000 m of the Southern Ocean has warmed since at least the 1930s, with the largest warming concentrated in

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the ACC. Böning et al. (2008) compared Argo observations from 2003 to 2008 to a historical climatology along mean streamlines of the ACC and showed that the Southern Ocean has both warmed and freshened to depths of at least 2000 m (Figure 18.8; see also Helm et al., 2010 and Durack and Wijffels, 2010). The zonally integrated linear trend in 0–2000 m ocean heat content (1955–1959 to 2006–2010) reaches a maximum between 40°S and 50°S (Levitus et al., 2012). Warming of the Southern Ocean below 1000 m accounts for about 10% of the increase in global ocean 0–700 m heat content between 1993 and 2003, or about 1 mm year^{-1} of local sea level rise, indicating both

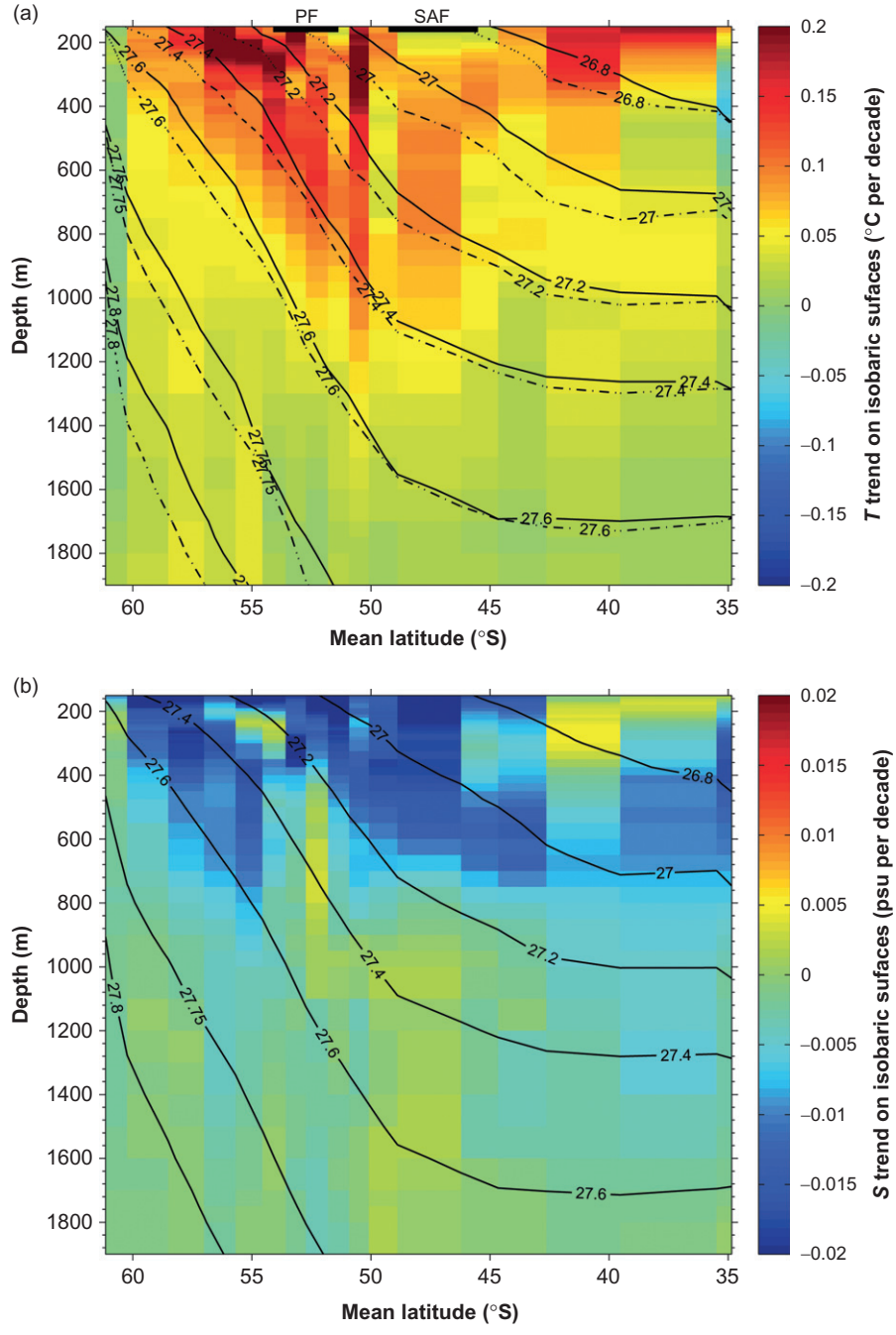

FIGURE 18.8 Temperature and salinity changes across the ACC, estimated by taking the difference between recent Argo measurements and a long-term climatology along mean streamlines. Mean decadal trends on pressure surfaces of potential temperature (a) and salinity (b). Black contours in (a) illustrate the migration of isopycnal surfaces during the past four decades: continuous (dashed) curves represent potential densities obtained by subtracting (adding) the linear trends over two decades from (to) the climatological mean density field. The position of the Polar Front (PF) and Subantarctic Front (SAF) are indicated in (a). *From Böning et al. (2008).*

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the deep-reaching nature of temperature changes in the Southern Ocean and their importance to global ocean energy budgets (Purkey and Johnson, 2010).

The warming of the Southern Ocean has been linked to a southward shift of the ACC, likely caused by a southward shift of the westerly winds (e.g., Böning et al., 2008; Gille, 2008). The wind changes have, in turn, been associated with a decadal-scale trend in the Southern Annular Mode (SAM), the dominant mode of variability in the southern hemisphere atmospheric circulation (e.g., Thompson et al., 2011). An increasing trend in sea surface height measured by satellite altimetry since 1993 is also consistent with an overall southward migration of the ACC (Figure 18.9). SSH contours corresponding to each of the ACC fronts migrated southward by about 0.6° of latitude between 1992 and 2007, but with significant regional variations (Sokolov and Rintoul, 2009a,b). Meijers et al. (2011) demonstrated that both adiabatic shifts of the ACC fronts and diabatic water mass changes driven by surface warming and an increase in precipitation–evaporation have contributed to the observed changes in temperature and salinity in the Southern Ocean. The inference of strong freshening driven by increased precipitation–evaporation in the ACC belt is consistent with the global analyses of salinity changes by Helm et al. (2010) and Durack and Wijffels (2010), who concluded that freshening of the Southern Ocean was the regional expression of a global amplification of the water cycle induced by the increased water vapor content of a warmer atmosphere.

5.2. Changes in the Southern Ocean Inventory of Dissolved Gases

The Southern Ocean inventory of carbon, oxygen, and chlorofluorocarbons (CFCs) has also changed in recent decades. As a result of the vigorous and deep-reaching overturning circulation, the Southern Ocean makes an important contribution to the global carbon cycle (e.g., Sabine et al., 2004). More than 40% of the ocean inventory of anthropogenic CO_2 entered the ocean south of 40°S (Khatiwala et al., 2013), mostly via the upper cell of the overturning circulation. More specifically, anthropogenic CO_2 is carried into the ocean interior through localized subduction windows, where Ekman pumping, eddy flux convergence, and the correlation between changes in mixed layer depth and the mean flow (lateral induction) conspire to transfer fluid across the base of the winter mixed layer (Sallée et al., 2012).

The Southern Ocean is the only region of the ocean where deep waters upwell to the sea surface and can exchange properties, including CO_2, with the atmosphere. While orbital changes determine the timing of glacial–interglacial cycles, carbon dioxide released to the atmosphere by upwelling of carbon-rich deep waters in the Southern Ocean provides the amplifier needed to complete the transition from glacial to interglacial conditions (e.g., Sigman et al., 2010). The effectiveness with which the Southern Ocean stores carbon dioxide depends on the response of both the natural and anthropogenic carbon cycles to changes in forcing. Le Quéré et al. (2007) argued on the basis of atmospheric measurements and an ocean circulation model that the Southern Ocean sink for CO_2 had saturated, due to increased outgassing of natural carbon resulting from a wind-driven increase in the overturning circulation in recent decades. The hypothesis sparked active debate (e.g., Law et al., 2007; Böning et al., 2008; Zickfeld et al., 2008) that helped focus attention on the question of how the overturning circulation responds to changes in wind forcing, or in other words, to what extent do eddies compensate wind-driven increases in Ekman transport? (See Section 4.)

Observational evidence of an increase in strength of the Southern Ocean overturning circulation has been difficult

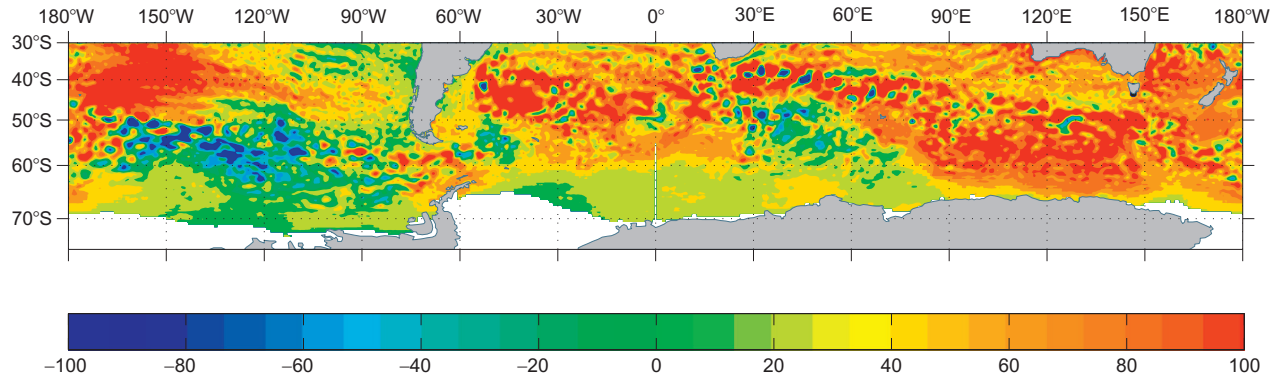

FIGURE 18.9 Change in sea surface height (SSH, in mm) between 1992 and 2012 based on satellite altimetry data (Collecte, Localization, Satellites (CLS)/Archiving, Validation, and Interpretation of Satellite Oceanographic Data (AVISO) "Mean Sea Level Anomaly" (MSLA), Le Traon et al. (1998)). SSH has increased over much (but not all) of the Southern Ocean and the southern flank of the subtropical gyres (yellow to red colors), consistent with an overall poleward shift of the ACC and poleward expansion and intensification of the subtropical gyres. *Figure courtesy of Serguei Sokolov, CSIRO.*

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to obtain. A few recent studies provide some support for the hypothesis that the overturning circulation has increased in recent decades. While data are sparse, Takahashi et al. (2012) show that the pCO_2 of Southern Ocean winter surface waters is increasing more rapidly than it is in the atmosphere, consistent with stronger upwelling of carbon from below. Changes in the inventory of CFCs along repeat hydrographic sections also suggest that the upper cell of the Southern Ocean overturning has intensified, with more rapid upwelling of deep water and greater formation and subduction of intermediate waters (Figure 18.10; Waugh et al., 2013). These observational studies are thus at least broadly consistent with the modeling and theoretical results discussed in Section 4, which indicate that eddies likely compensate some, but not all, of the larger Ekman transport in response to strengthening winds. On the other hand, Helm et al. (2011) interpret a decline in oxygen between the 1970s and 1990s in AAIW and SAMW as evidence for reduced ventilation as a result of enhanced stratification.

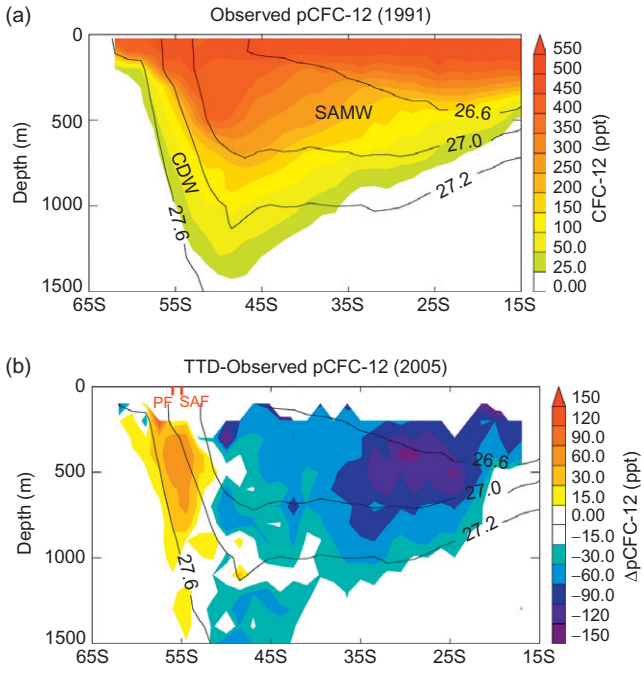

FIGURE 18.10 (a) Observed chlorofluorocarbon-12 concentration (pCFC-12) in 1991 along a north–south transect at 150°W in the Pacific. (b) Difference between modeled and observed pCFC-12 in 2005, where the modeled value is based on a transit time distribution approach assuming steady transport. Positive (negative) values indicate modeled pCFC-12 values are higher (lower) than observed. The pattern of differences is consistent with an increase in the overturning circulation, with stronger sinking of well ventilated intermediate waters north of the ACC and stronger upwelling of poorly ventilated deep water south of the ACC. Black lines show the potential density distribution in 1991. CDW, Circumpolar Deep Water; SAMW, Subantarctic Mode Water; PF, Polar Front; SAF, Subantarctic Front. *From Waugh et al. (2013).*

5.3. Changes in Southern Ocean Water Masses

As water mass properties are largely set at the sea surface through air–sea interaction, and then transferred into the ocean interior by subduction, analysis of changes in water mass properties can provide clues about changes in surface climate. Water masses also influence the interior stratification and distribution of PV, so water mass changes can have dynamical implications.

5.3.1. Changes in Mode and Intermediate Waters

Coherent changes in the properties of the water masses involved in the upper cell (SAMW and AAIW) have been observed in recent decades. SAMW has generally cooled and freshened (e.g., Böning et al., 2008; Helm et al., 2010). The AAIW salinity minimum core shoaled and warmed between 1970 and 2009, with regionally varying trends in salinity (Schmidtko and Johnson, 2012). Both an increase in precipitation–evaporation and a warming-induced poleward migration of isopycnal outcrops (i.e., into regions of higher or lower surface salinity) have likely contributed to the observed trends in intermediate water properties (Durack and Wijffels, 2010; Helm et al., 2010). AAIW changes in particular locations have been linked to other processes, including exchange between the Indian and Atlantic basins (McCarthy et al., 2011) and changes in surface forcing related to modes of climate variability (Naveira Garabato et al., 2009).

5.3.2. Changes in AABW

Perhaps the most dramatic water mass changes observed in the Southern Ocean are those of the AABW. AABW has warmed, freshened, decreased in density, and contracted over the last 40 years (e.g., Jacobs et al., 2002; Rintoul, 2007; Jacobs and Giulivi, 2010; Purkey and Johnson, 2010, 2012, 2013; Fahrbach et al., 2011; Shimada et al., 2012; van Wijk and Rintoul, 2013; see also Chapter 19). Basin-average warming is largest near Antarctica and decreases with distance along the AABW export pathways (Purkey and Johnson, 2010). Globally, the volume of water cooler than 0 °C (a common definition of AABW) has decreased by 8.2 ± 2.6 Sv over the last two decades (Purkey and Johnson, 2012); in the Australian Antarctic Basin, the layer of AABW denser than the 28.3 kg m^{-3} neutral density surface has thinned by more than 100 m decade^{-1} and lost more than half its volume since 1970 (van Wijk and Rintoul, 2013). The strongest freshening signal is found in the Indian and Pacific basins, where both the Ross Sea and Adélie Land sources of bottom water have freshened (Jacobs et al., 2002; Aoki et al., 2005; Rintoul, 2007;

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Jacobs and Giulivi, 2010; Purkey and Johnson, 2013). Freshening of Ross Sea shelf waters has been attributed to inflow of glacial ice melt in the Amundsen–Bellingshausen Seas, although changes in sea ice formation and melt or precipitation may also contribute (Jacobs and Giulivi, 2010). van Wijk and Rintoul (2013) argue that freshening of the dense source waters is the primary cause of the observed changes in AABW, based on the evolution of water mass properties along the boundary currents exporting AABW to the north in the Australian Antarctic Basin. Near the sources, strong freshening is observed, with little change in temperature. Salinity changes decrease downstream, while temperature changes increase, consistent with gradual warming by enhanced mixing along the export pathway due to freshening-induced changes in abyssal stratification (e.g., Shimada et al., 2012). The oxygen content of the densest waters remained constant within measurement error between the mid-1990s and 2012, suggesting that the contraction of the AABW layer results from formation of slightly less dense varieties of AABW rather than a reduction in the formation rate.

5.4. Ocean–Ice Shelf Interaction

Enhanced basal melt of floating ice shelves has been identified as a likely source of additional freshwater input to the Southern Ocean (Shepherd et al., 2004; Rignot et al., 2011). Thinning of the Pine Island Glacier in the eastern Pacific sector of the Southern Ocean has been particularly dramatic. Recent measurements of the sub-ice shelf cavity using the autonomous underwater vehicle Autosub have revealed that water with temperatures 4 °C above the *in situ* freezing point reaches the grounding line, driving rapid melt (Jenkins et al., 2010; Jacobs et al., 2011). Pritchard et al. (2012) concluded from satellite measurements that 20 of Antarctica's 54 ice shelves and glacier tongues thinned during the short period from 2003 to 2008 as a result of enhanced basal melt. Because ice shelves buttress the interior ice sheet, thinning or collapse of the floating ice shelves can cause more rapid loss of grounded ice and hence more rapid rise in sea level.

The oceanic heat flux toward the Antarctic margin is therefore a key factor controlling the mass balance of the Antarctic ice sheet. Changes in either temperature or circulation could alter the ocean heat flux to the base of the ice shelf. More specifically, warm ocean waters need to cross a number of barriers to reach the base of the floating ice: the shelf break, the continental shelf, the ice front, and the boundary layer beneath the ice. The dynamics controlling the net ocean heat flux to the base of the ice are therefore complex and likely to depend on large-scale forcing (e.g., wind stress, sea ice, buoyancy forcing) and circulation (e.g., proximity to the ACC), as well as small-scale (e.g., turbulent mixing rates) and local factors (e.g., topography of the sea floor and the ice shelf base). Measurements from beneath the Pine Island Glacier underscore this point: basal melt rates likely increased when the glacier thinned enough to float free of a submarine ridge, allowing stronger inflow of relatively warm ocean waters (Jenkins et al., 2010). This example highlights the difficulty of assessing the sensitivity of the Antarctic ice sheet mass balance to warming of the Southern Ocean. The situation is made even more challenging by the fact that very few observations of temperature and velocity have been made on the Antarctic continental shelf, and modeling studies are hampered by lack of knowledge of even such basic features as the bathymetry.

5.5. Changes in Southern Ocean Sea Ice

Satellite data suggest that there has been a small increase in the extent of Southern Ocean sea ice (by about 1% per decade between 1978 and 2010; Parkinson and Cavalieri, 2012), in stark contrast to the Arctic. The small overall increase is the sum of strong trends of opposite sign in the Bellingshausen–Amundsen and Ross Seas. The retreat of sea ice in the Bellingshausen–Amundsen seas has extended the summer ice-free period by more than 3 months since the late 1970s, a change comparable to that observed in the Arctic; expansion of sea ice in the Ross Sea has reduced the ice-free period by 2 months over the same time period (e.g., Stammerjohn et al., 2012). These regional changes in the extent and duration of the Southern Ocean sea ice can largely be explained by trends in surface winds, through both sea ice advection and changes in ice formation and melt rates (Holland and Kwok, 2012). Input of glacial meltwater may have contributed to the expansion of Southern Ocean sea ice by enhancing near-surface ocean stratification and restricting heat input from below (Hellmer, 2004; Bintanja et al., 2013). The changing extent and seasonality of sea ice over broad regions of the Southern Ocean is likely to alter the surface stress and buoyancy fluxes that drive the subpolar circulation, but these effects have yet to be quantified.

5.6. Causes of Recent Southern Ocean Change

As in many other parts of the ocean, changes in recent decades are likely to be associated, directly or indirectly, with changes in wind forcing. Many studies have linked observed changes in the Southern Ocean to the trend in the westerly winds associated with the SAM (e.g., Hall and Visbeck, 2002; Thompson et al., 2011), including the southward shift of the ACC and resulting changes in temperature and salinity (e.g., Gille, 2008), warming and freshening of shelf waters (e.g., Jacobs and Giulivi, 2010), and

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changes in ocean carbon uptake (e.g., Le Quéré et al., 2007), among many others. The schematic ocean response to an intensification of winds (e.g., Hall and Visbeck, 2002) is straightforward to understand: stronger westerly winds mean more equatorward Ekman transport, with enhanced upwelling to the south and enhanced downwelling to the north; a southward shift of the westerlies will drive a southward shift of the ACC; and stronger northward Ekman transport might drive an expansion of the Southern Ocean sea ice. However, in many cases a quantitative understanding of the link between changes in winds (i.e., the SAM) and ocean change is lacking, in part due to a scarcity of observations. The oceanographic literature has not always taken the seasonal (strongest trend in summer) and longer term variations of the SAM (the time history is not a simple monotonic trend in time) into account. A key question, once again, is the response of the overturning circulation to a change in wind forcing. Schematics like that of Hall and Visbeck (2002) are based on the assumption that eddies do not compensate Ekman transport. In addition, while most studies have focused on the effect of changes in the westerlies, changes in meridional winds over the Southern Ocean are likely to be important for many oceanographic phenomena such as air–sea heat flux and the depth of the surface mixed layer (e.g., Sallée et al., 2010b), and sea ice extent (e.g., Turner et al., 2009; Holland and Kwok, 2012). Warming of the surface layer and an increase in precipitation and ice melt have likely changed the buoyancy forcing of the Southern Ocean, with likely impacts on the circulation, but these effects have not yet been quantified.

6. SUMMARY AND OUTSTANDING CHALLENGES

A common theme to emerge from the observational, theoretical, and numerical studies reviewed here is the significance of departures from zonal symmetry. Topography steers the mean flow and localizes many of the dynamical interactions that govern the circulation of the Southern Ocean and its sensitivity to changes in climatic forcing, including ACC instability, EKE, eddy stirring and meridional exchange, interfacial and bottom form stress, and diapycnal mixing through the generation and breaking of internal waves. Another example of the localization of important dynamical processes is the generation of narrow subduction windows by changes in mixed layer depth along mean streamlines (Figure 18.4). Thus, while the key ingredients in the dynamical recipe for the Southern Ocean circulation have arguably been identified, it is likely that further progress will require disentangling the details of the dynamics of local regions that dominate the circumpolar budgets of momentum, vorticity, and other dynamical quantities. In particular, the climatic sensitivity of the overturning circulation is intrinsically linked to eddy–mean flow–topography interactions in specific areas, which influence the response of upper-ocean diapycnal mixing to wind and eddy effects and the resilience of the Southern Ocean PV structure.

Changes in the Southern Ocean overturning circulation may feed back on climate by altering the capacity of the ocean to take up heat and carbon dioxide or by affecting the stability of the Antarctic cryosphere. Therefore, improved understanding of the response of the Southern Ocean overturning to wind and buoyancy forcing perturbations remains a high priority. While a consensus seems to be emerging from models and theory that the Southern Ocean circulation approaches an eddy-saturated and partially eddy-compensated state, the details of how this state is maintained are not yet clear.

The observational record is often too short to provide clear evidence to support or refute specific hypotheses concerning the response of the Southern Ocean circulation to changes in forcing. However, evidence is growing that changes are underway in the region that are of potential significance to climate, biogeochemical cycles, and the mass balance of the Antarctic ice sheet. Observations are particularly sparse in the deep ocean and within the sea ice zone. The need to better understand the nature, causes, and consequences of Southern Ocean change provides strong motivation for enhancements to the Southern Ocean observing system.

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